

# Week 15: Bonus: monad transformers

Functional Programming in Lean

# Learning goals

- ▶ Read a transformer stack as a result type
- ▶ Compare two orders of state and error
- ▶ Predict whether state survives failure

# Central idea

The order of effects changes observable results.

# Lean example

```
abbrev LoseState :=  
  StateT Nat (Except String)  
abbrev KeepState :=  
  ExceptT String (StateM Nat)
```

# What to notice

- ▶ The first stack may lose the final state
- ▶ The second returns state alongside an error
- ▶ A shared evaluator can use an effect interface

# Live coding

1. Complete both failing computations
2. Run each from state zero
3. Explain the different outputs by expanding the types

# Pause and predict

Which stack can return a final state after an error?

# Exercise connection

Open `exercises/Week15.md` after the lecture.

The complete example is `Course/Week15.lean`; the fill-in version is `Templates/Week15.lean`.